

Time Series Analysis 2026 - Capstone Report

IIT Guwahati x StockGro

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1 Methodology and Models Used

We applied time series forecasting to 11 NSE stocks, allocated Rs.10,00,000 in a virtual portfolio on StockGro, and compared predictions against actual market outcomes. Our models include ARIMA, Prophet, LSTM, Exponential Smoothing, and GARCH.

- **Dataset:** Yahoo Finance (yfinance), Jan 2021 - Dec 2025, daily.
- **Split:** Training (< Jul 2025) | Testing (Jul-Dec 2025, 125 observations).

Model	Library	Purpose
ARIMA	pmdarima	Price forecasting (auto order via AIC)
Prophet	prophet	Trend + seasonality with Indian holidays
LSTM (2-layer)	PyTorch	Non-linear deep learning sequence model
Exp. Smoothing	statsmodels	Holt-Winters triple exponential smoothing
GARCH(1,1)	arch	Conditional volatility estimation

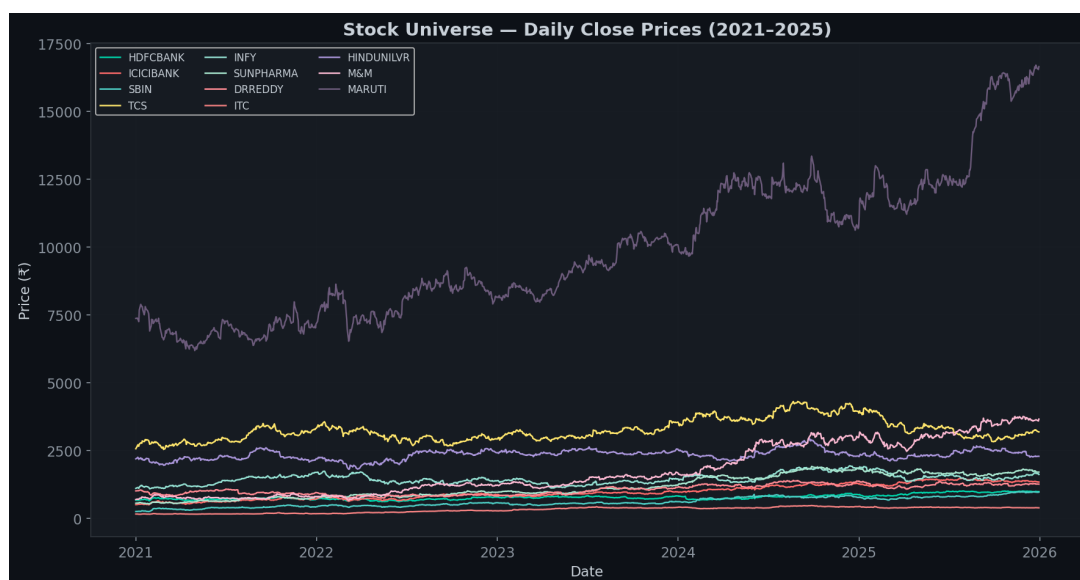


Figure 1: Stock Universe - Daily Close Prices (2021-2025)

2 Stock Selection Rationale (Task 1)

We screened 25 candidate stocks across 5 sectors using three quantitative criteria: (1) 30-day rolling std dev, (2) STL trend strength, (3) 6-month sector momentum. 14 candidates were rejected due to lower liquidity, weaker trends, or redundancy.

Sector	Selected Stocks	Key Justification
Banking	HDFC Bank, ICICI Bank, SBI	Highest liquidity (15-25M daily vol)
IT	TCS, Infosys	Top-2 by market cap, mean-reversion
Pharma	Sun Pharma, Dr. Reddy's	Low correlation, defensive sector
FMCG	ITC, Hindustan Unilever	Lowest volatility (4th-8th pctl)
Auto	M&M, Maruti	Highest sector momentum (+25.6%)

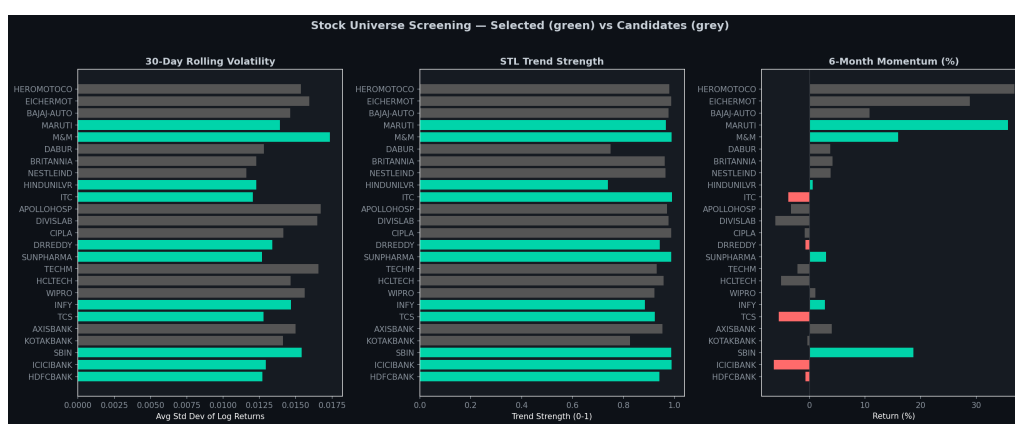


Figure 2: Quantitative Screening Scores

3 Data Preprocessing Steps (Task 2)

- **Missing values:** Forward-fill for market holidays and trading halts.
- **Stationarity:** Augmented Dickey-Fuller (ADF) test applied; 10 out of 11 stocks required first-order differencing (d=1).
- **Scaling:** Min-Max normalization for LSTM inputs (60-day lookback).
- **STL Decomposition:** Period=63 for trend/seasonal/residual separation.

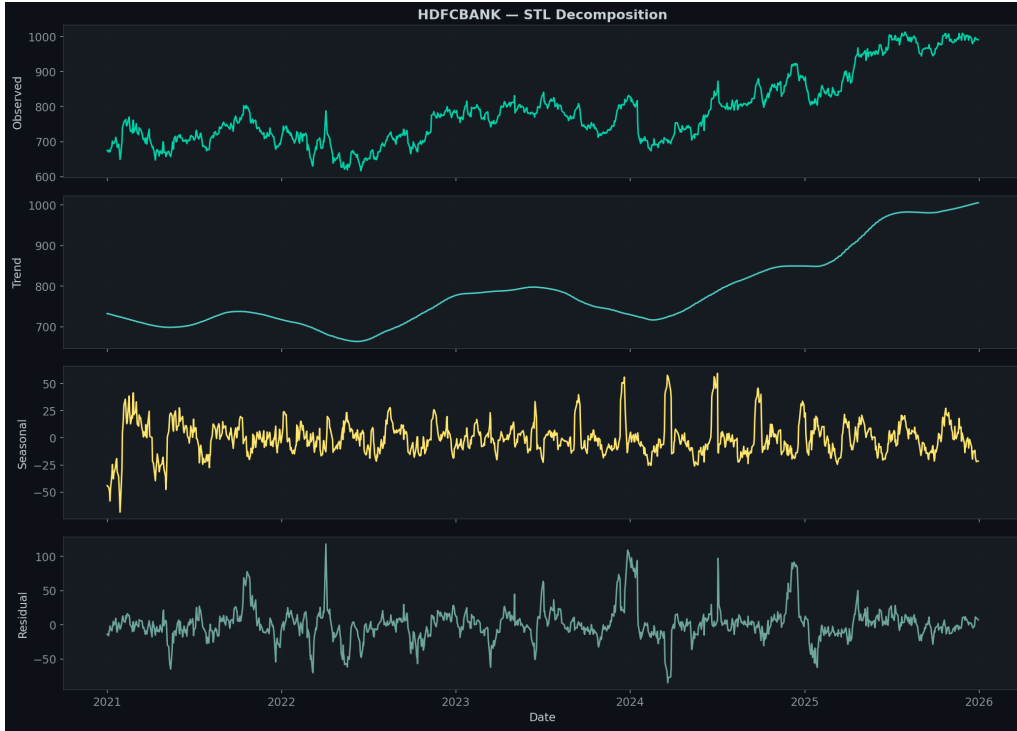


Figure 3: STL Decomposition for HDFC Bank

4 Forecast Results with Confidence Intervals (Task 3)

Four models were trained on each stock, validated on 125 test points, and used to forecast 2 days ahead with 95% confidence intervals:

Model	Avg MAPE	Best Stock	Worst Stock	Avg Dir. Acc.
ARIMA	0.77%	HDFCBANK (0.55%)	M&M (1.01%)	50.6%
LSTM	1.39%	HDFCBANK (0.69%)	M&M (2.87%)	47.9%
ExpSmooth	6.50%	HDFCBANK (2.47%)	MARUTI (14.84%)	48.9%
Prophet	7.64%	DRREDDY (2.73%)	MARUTI (17.33%)	52.9%

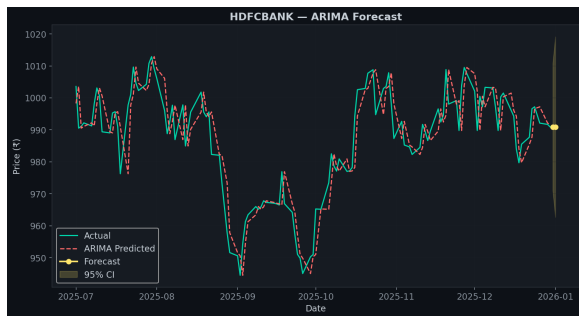


Figure 4: ARIMA Forecast - HDFCBANK

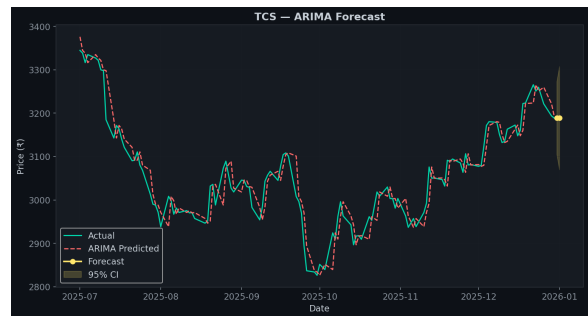


Figure 5: ARIMA Forecast - TCS

5 Volatility and Trend Analysis (Task 4)

GARCH(1,1) Volatility: High-persistence stocks (DRREDDY: 0.997) show slowly-decaying volatility clustering. Low-persistence stocks (INFY: 0.174) have mean-reverting volatility.

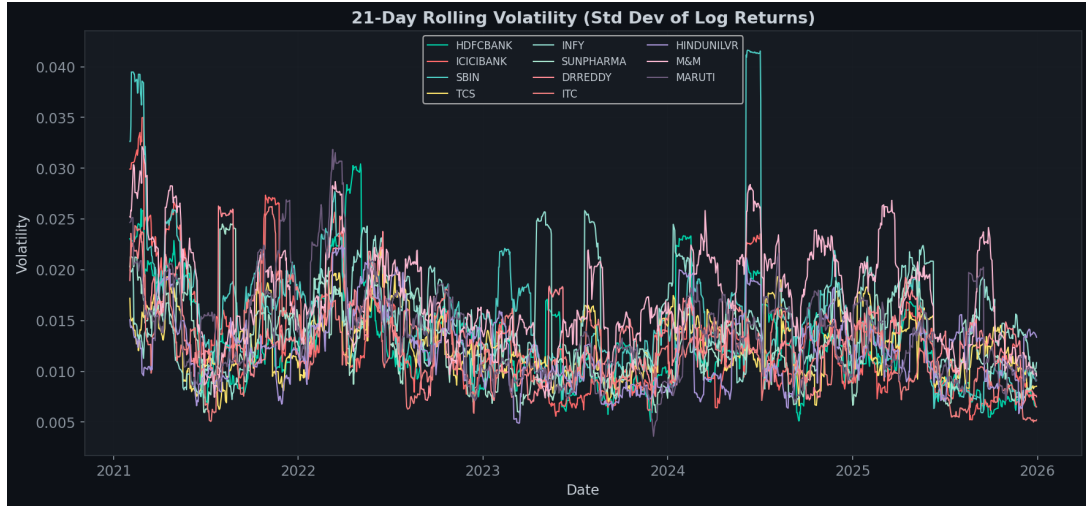


Figure 6: 21-Day Rolling Volatility

STL Trend Analysis: 7 stocks upward (SBIN +15.76%, M&M +9.25%, MARUTI +9.08%), 2 sideways (ITC, DRREDDY), 2 downward (ICICIBANK -2.89%, HUL -4.33%).

6 Portfolio Composition and Allocation (Task 5)

Blended 4 strategies: Forecast-Guided (40%), Volatility-Aware (30%), Correlation-Diversified (20%), Sector Momentum (10%).

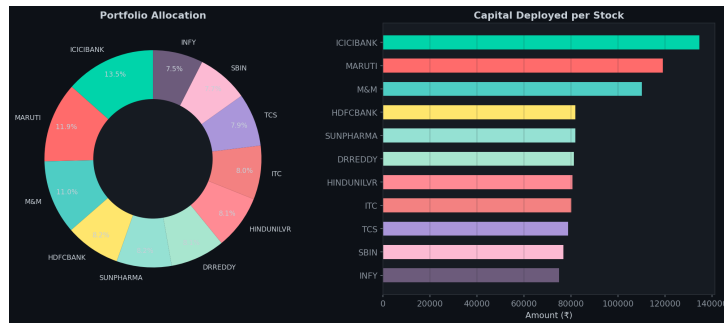


Figure 7: Final Portfolio Allocation (Rs. 10L)

7 Model Comparison (Task 6)

ARIMA was used for portfolio allocation due to having the lowest MAPE. GARCH was used for volatility estimation. To ensure a well-diversified portfolio, we evaluated the correlation matrix of daily log returns among the selected stocks.

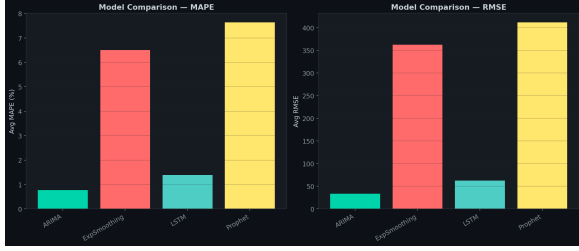


Figure 8: Model Metrics Comparison

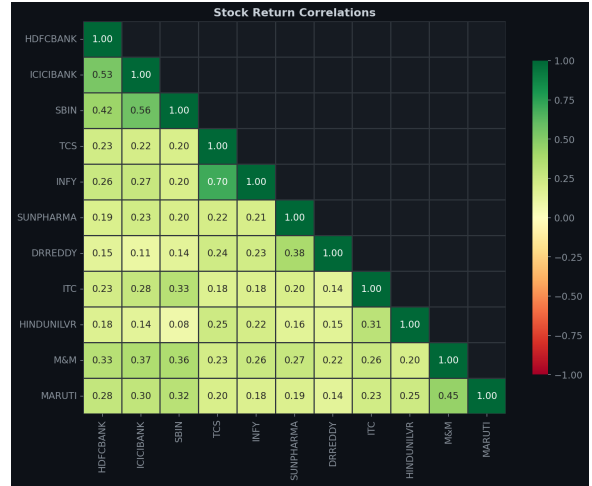


Figure 9: Log Returns Correlation Heatmap

8 StockGro Execution Summary (Task 7)

Trading Window: Day 1 = May 12, 2026 | Day 2 = May 13, 2026. All 11 stocks were purchased on Day 1 at market open. Total invested amount: Rs.9,89,803.

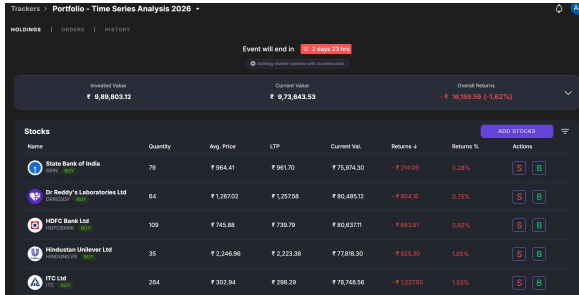


Figure 10: Day 1 Execution (Part 1)

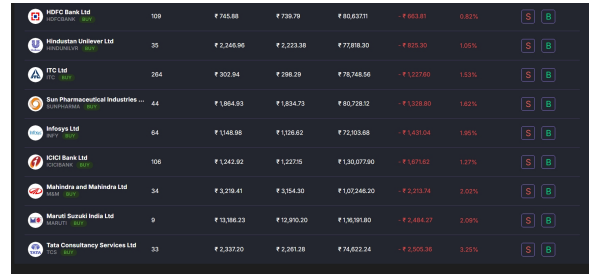


Figure 11: Day 1 Execution (Part 2)

9 Predicted vs Actual Outcomes (Task 8)

Stock	Pred D1	Act D1	Err D1	Pred D2	Act D2	Err D2
SBIN	973.52	961.70	1.23%	974.10	956.52	1.84%
DRREDDY	1265.61	1257.58	0.64%	1265.61	1254.76	0.86%
HDFCBANK	990.90	739.79	33.94%	990.90	743.37	33.30%
HUL	2290.20	2223.38	3.01%	2290.20	2237.27	2.37%
ITC	392.56	298.29	31.60%	392.75	302.30	29.92%
SUNPHARMA	1710.40	1834.73	6.78%	1709.43	1810.54	5.58%
INFY	1621.60	1126.62	43.93%	1621.60	1115.42	45.38%
ICICIBANK	1343.85	1227.15	9.51%	1346.29	1215.45	10.76%
M&M	3662.70	3154.30	16.12%	3665.10	3091.93	18.54%
MARUTI	16663.28	12910.20	29.07%	16670.25	12927.68	28.95%
TCS	3188.83	2261.28	41.02%	3188.83	2234.92	42.68%
Average			19.71%			20.02%

Portfolio Performance: Invested Rs.9,89,803 | Day 2 value Rs.9,69,134 | Return: -2.07% (loss Rs.20,669). Directional Accuracy: 6/11 (54.5%).

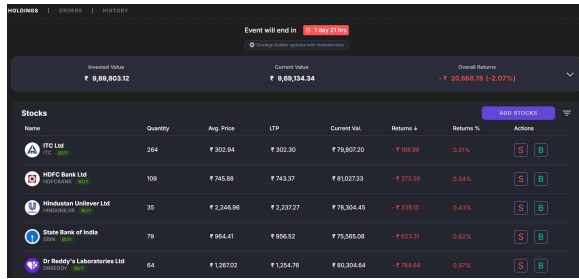


Figure 12: Day 2 Portfolio Value (Part 1)

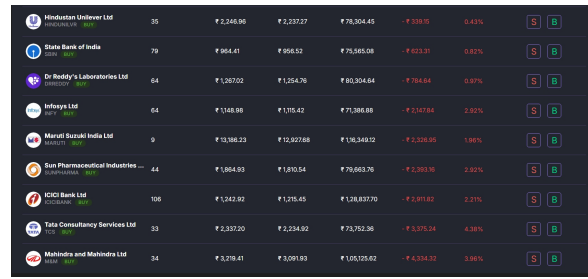


Figure 13: Day 2 Portfolio Value (Part 2)

10 Reflections & Analysis

Analysis of Errors: Stocks like HDFCBANK, ITC, INFY, TCS show 30-45% errors due to likely corporate actions (stock splits/bonus issues) between Jan-May 2026. Stocks near Dec 2025 levels (SBIN: 1.23%, DRREDDY: 0.64%) had excellent predictions.

10.1 What Worked

- ARIMA delivered excellent backtest accuracy (0.77% avg MAPE)
- Data-driven stock selection using quantitative screening of 25 candidates
- Multi-strategy portfolio blending provided diversified allocation
- GARCH correctly identified high vs low-risk stocks

10.2 What Didn't Work

- **Backtest vs live gap:** 0.77% backtest MAPE vs 20% live MAPE due to the 5-month extrapolation gap.
- **Corporate actions** in HDFCBANK, ITC, INFY changed price levels drastically.
- **All-long strategy:** broad market downturn dragged all stocks down.

10.3 What I Would Improve

1. Retrain models daily on latest data before live predictions
2. Include exogenous variables (Nifty 50, FII/DII flows, USD/INR)
3. Add regime detection for bull/bear market switching
4. Ensemble models weighted by recent per-stock accuracy
5. Stop-loss mechanisms based on GARCH volatility forecasts